case mapping – the rules for mapping characters when converting case. I promise this will make more sense when we examine the `mb_convert_case()` function (which is the underlying implementation for `mb_strtoupper()` and its counterpart) below.

ucfirst(), ucwords(), lcfirst()

A pair of simple functions, and like `strtolower()`, they only act on the ASCII alphabet (a–z, bytes *0x61* to *0x7a*). In the case of `ucwords()`, every word in the string is affected, where a word is considered to be any set of bytes that is not a separator byte. Default separators are *0x09* (horizontal tab), *0x0A* (new-line), *0x0B* (vertical-tab), *0x0C* (new-page), *0x0D* (carriage-return), and *0x20* (space) – but can be overridden by supplying a string of bytes as a second argument if required.

```
print ucfirst('only affects the very first letter. nothing else.');
print PHP_EOL;
print ucwords('every first letter of eVERY wORD.');
```

```
Output:
Only affects the very first letter. nothing else.
Every First Letter Of EVERY WORD.
```

The results of the function calls are self-explanatory. Although not shown here, the function `lcfirst()` exists and does the inverse of `ucfirst()`. I'm sure there's a use-case for it… In previous versions of PHP, these functions were more sophisticated and would vary their behavior with locale settings. Since PHP 8.2 they only work on the basic ASCII Latin alphabet. For more robust case conversion and multi-byte safety, we have a better option – `mb_convert_case()` – which we'll get to very shortly.

mb_ucfirst(), mb_lcfirst()

A new feature for PHP 8.4, `mb_ucfirst()` and `mb_lcfirst()` are the multi-byte safe equivalents to `ucfirst()` and `lcfirst()`. Whereas these core functions only process bytes in the ASCII range (and will cause havoc for non-UTF-8 multibyte data), the `mbstring` equivalents are both safe to use on any (supported) multibyte encoding and are Unicode-aware.

```
$lcname = "\u{142}ukasz";
$ucname = "\u{141}UKASZ";
```